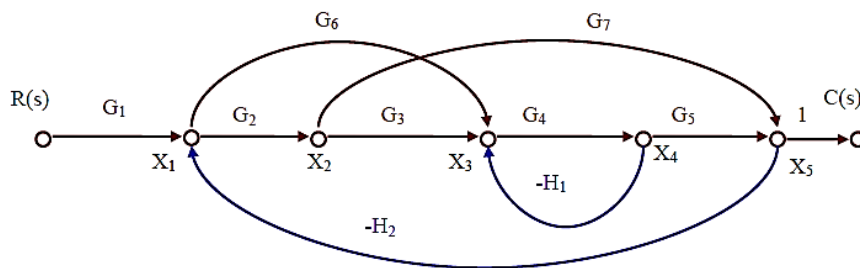


Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Obtain the Transfer function of the given block diagram	Level-5	6
B) Derive Transfer function and Mathematical Model of Mechanical System	Level-2	6
C) Obtain Transfer function of C(s)/R(s) using Masson's gain formula for the SFG shown	Level-5	6



Q.2 Solve Any Two of the following. **12**

- A) find the steady state error for an input signal Level-5 **6**

$$r(t) = \left(5 + 2t + \frac{t^2}{2}\right) u(t)$$

of unity negative negative feedback control system with

$$G(s) = \frac{5(s+4)}{s^2(s+1)(s+20)}$$

- B) find the stability of the control system having characteristic equation Level-4 **6**

$$s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$$

- | | | |
|---|----------------|----------|
| C) What do you mean by time domain specifications?
Explain all parameters used in details. | Level-3 | 6 |
|---|----------------|----------|

Q. 3 Solve Any Two of the following.		12
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- | | | |
|--|----------------|----------|
| A) With suitable block diagrams and equations, explain the following types of controllers employed in control systems: (Any two) | Level-2 | 6 |
|--|----------------|----------|

- a) Proportional controller,
- b) Proportional-plus-integral controller,
- c) PID controller,
- d) Integral controller

- | | | |
|---|----------------|----------|
| B) find the time domain specifications of a control system having the closed loop transfer function | Level-4 | 6 |
|---|----------------|----------|

$$\frac{4}{s^2 + 2s + 4}$$

when the unit step signal is applied as an input to this control system.

- | | | |
|---|-------------------|----------|
| C) While applying Routh's Stability Criterion what difficulties arise causing the breakdown of Routh's test. How these difficulties overcome? | Level-3, 2 | 6 |
|---|-------------------|----------|

Q.4 Solve Any Two of the following.		12
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- | | | |
|---|------------------|----------|
| A) Comment on Stability by drawing Bode Plot For given OLTF . | Level - 4 | 6 |
|---|------------------|----------|

$$\frac{K(1 + 0.2S)(1 + 0.025S)}{S^3(1 + 0.001S)(1 + 0.005S)}$$

- | | | |
|---|------------------|----------|
| B) The OLTF of Unity feedback system is given by, | Level - 4 | 6 |
|---|------------------|----------|

$$G(S) = \frac{K}{(S + 3)(S + 5)(S^2 + 2S + 2)}$$

In the Root Locus diagram for this system,

- i) Determine no. of branches of root loci and
- ii) Determine location at which these branches starts and terminates
- iii) Find the angle of asymptote and the point at which asymptotes intersect the real axis
- iv) Find the breakaway point and the point where root loci cross the imaginary axis

- | | | |
|---|------------------|----------|
| C) Explain the following characteristics of feedback control system | Level - 2 | 6 |
|---|------------------|----------|

- Stability
- Disturbance rejection
- Insensitivity

Q. 5 Solve Any Two of the following.**12**

- A)** Consider the open loop transfer function of a closed loop control system

Level - 5**6**

$$G(s)H(s) = \frac{5}{s(s+1)(s+2)}$$

draw the polar plot for this control system

- B)** Find the state space model for the system having transfer function.

Level - 5**6**

$$\frac{Y(s)}{U(s)} = \frac{1}{s^2 + s + 1}$$

- C)** Verify the controllability and observability of a control system which is represented in the state space model as

Level - 5**6**

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} [u]$$

$$Y = [0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

***** End *****